ASPECT WAVE LLC

Vessel Measurement Recorder

November 2026 · Prepared by Sophie Scopazzi

Bill of Materials

No custom circuit boards are required. Every part connects either by stacking Feather boards on top of each other or a shared I2C cable chain (STEMMA QT / Qwiic).

Function	Part	Num	Cost
Processor and Wi-Fi	Adafruit ESP32-S2 Feather w/ w.FL to SMA adapter and external Wi-Fi antenna	6282, 5444, 945	$29.5
Position	Ultimate GPS FeatherWing (PA1616D) w/ external active antenna	3133, 960	$47
Clock and storage	Adalogger FeatherWing (precision real-time clock w/ CR1220 and microSD)	2922, 380, 5252 / 1294	$15 / $33.5
Display	OLED FeatherWing (128 × 64) with buttons + reset button	4650	$15
Motion	BNO085 9-axis orientation sensor (I2C)	4754	$29.5
Persistent memory	24LC32 EEPROM (I2C)	5146	$4
Weather – Pressure	BMP388 pressure (I2C),	3966	$10
Temperature & humidity	SHT30 temperature and humidity connected w/ Qwiic / STEMMA QT breakout board (I2C)	5064, 5961	$11
GPIO Expansion Board	PCF8574 GPIO for mark and cast buttons (and extra ports)	5545	$5
ADC Expansion Board	ADS7830 ADC for dimmer pot	5836	$6
OLED + function LEDs power / dimmer	10k linear potentiometer with built-in switch	3395	$2
System power switch	Latching power switch	1476	$2
Screen & Mark button	Momentary pushbuttons (2x)	1439	$3
LED indicator lights	4 panel-mount 3.3 V LEDs, multicolor pack (red / green used)	Many vendors	$1
Wires	STEMMA QT 50 mm (x6) and arcade quick-connect wires	4399, 1152	$11
Terminal Blocks	Terminal blocks, 2.54 mm pitch: one 4-pin, two 8-pin	2137, 2141	$7
Power	2,500 mAh – 10,000 mAh LiPo battery with gauge w/ USB-C small round panel-mount extension cable	328 or 5035, 6069	$19.5 / $34.5
Housing	Weatherproof enclosure	Pelican 1040	$28
Miscellaneous parts for affixing and etc.	Double-sided Velcro, black nail polish (optional, to shade green always-on power LEDs on I2C boards)	Many vendors	$10

Function	Part	Num	Cost
	(full) **Vessel Measurement Recorder**	**$256 / $290**	
	(without weather) **Vessel Measurement Recorder**	**$235 / $269**	
	(inclinometer) **Vessel Motion Recorder**	**$173 / $188**	
	(no IMU, weather sensors only) **Weather Station**	**$203 / $239**	
	(no sensors except GPS) **Deployment Logger**	**$145 / $181**	